

Continuous renal replacement therapy principles

Rolando Claure-Del Granado MD, FASN^{1,2}  | William R. Clark MD³

¹Division of Nephrology, Hospital Obrero No 2 – CNS, Cochabamba, Bolivia

²Universidad Mayor de San Simon School of Medicine, Cochabamba, Bolivia

³Davidson School of Chemical Engineering, Purdue University, West Lafayette, IN, USA

Correspondence

Rolando Claure-Del Granado, MD, FASN, Hospital Obrero No 2 – CNS, Universidad Mayor de San Simon, School of Medicine, Cochabamba, Bolivia.
Email: rclaure@yahoo.com

Abstract

Continuous renal replacement therapy (CRRT) is an extracorporeal blood purification therapy that aims to support kidney function over an extended period of time. One of the main objectives of CRRT is the removal of excess fluid and solutes retained as a consequence of acute kidney injury. Because prescription of CRRT requires goals to be set with regard to the rate and extent of solute and fluid removal, a comprehensive understanding of the mechanism by which solute and fluid removal occurs during CRRT is essential. Basic mechanisms of fluid transport and solute removal (ultrafiltration, diffusion, convection, and adsorption) and the factors influencing these processes in CRRT are described. From the combination of the different transport mechanisms, a number of CRRT modalities are identified and described. Finally, these principles are applied to provide a brief overview of the concept of effluent-based CRRT dose.

1 | INTRODUCTION

Renal replacement therapy is frequently required in critically ill patients developing acute kidney injury (AKI).¹ Continuous renal replacement therapy (CRRT) is any extracorporeal blood purification therapy usually used in critically ill hemodynamically unstable patients in the intensive care unit and aims to support kidney function over an extended period of time. One of the primary objectives of CRRT is the removal of excess fluid and blood solutes that are retained as a consequence of decreased or absent glomerular filtration.

Because prescription of CRRT requires goals to be set with regard to the rate and extent of both solute and fluid removal, a comprehensive understanding of the mechanisms by which solute and fluid removal occurs during CRRT is important. The transport mechanisms operational in different CRRT modalities are discussed and these principles are applied to provide a brief overview of the concept of CRRT dose.

2 | MEMBRANE CHARACTERISTICS

Similar to a filter used for conventional hemodialysis, a CRRT filter is comprised of thousands of hollow fiber membranes. The

ultrafiltration coefficient (K_{UF}) is the measurement used clinically to quantify the water permeability of the filter. Technically, K_{UF} is estimated from the relationship between ultrafiltration rate (Q_{UF}) versus transmembrane pressure (TMP) line and is obtained from experiments utilizing blood at a specific hematocrit as the test solution. Depending on several factors, this curve is linear until developing a Q_{UF} plateau at a certain TMP. K_{UF} (ml/h/mm Hg) is the slope of the linear phase of the curve. Transmembrane pressure is estimated using the following formula:

$$TMP = \left(\frac{P_{PRE} + P_{OUT}}{2} \right) - P_{EFF}, \quad (1)$$

where P_{PRE} is the pre-filter pressure, P_{OUT} the post-filter pressure, and P_{EFF} the pressure measured in the effluent line (all three typically measured by the machine).

As mentioned above, K_{UF} values provided by manufacturers are derived from experiments with blood. Blood components substantially influence the effective water permeability, which may differ markedly from the inherent water permeability (i.e., that obtained for a “virgin” membrane). **Fouling** (also known as secondary membrane formation), the non-specific adsorption of plasma proteins to a membrane, is the most important determinant of effective membrane permeability. The mechanisms involved in the fouling of

extracorporeal membranes are not precisely defined although proteins over a wide range of molecular weights participate. Plasma proteins having a molecular size that allows some degree of membrane pore entry (generally, molecular weight <50 kDa) may adsorb to pore walls, effectively resulting in pore constriction. Proteins completely or near completely excluded from pores (i.e., molecular weight 70 kDa or more) can foul the membrane's nominal surface, resulting in pore "blockage" and elimination of a certain percentage of pores with respect to participation in solute removal. While the first adsorptive mechanism may actually account for the beneficial removal of some toxins (e.g., low-molecular weight toxins; see below), it is the second mechanism that mediates membrane fouling in the classical sense.

K_{UF} is still used commonly in chronic hemodialysis to categorize "high-flux" versus "low-flux" dialyzers. However, differentiating CRRT filters in this manner has little relevance since the vast majority of membranes comprising CRRT filters are highly permeable (i.e., large average pore size). Nevertheless, a K_{UF} of 25 ml/h/mmHg is a reasonable threshold defining a high-flux CRRT filter.

3 | MECHANISM OF SOLUTE AND FLUID TRANSPORT

3.1 | Solute removal

Solute removal in CRRT can occur by different mechanisms that are described below:

3.1.1 | Diffusion

Diffusion is a process whereby molecules move randomly across a semipermeable membrane. Solute movement occurs from a more concentrated to a less concentrated area, until an equilibrium is reached between the two compartments (Figure 1). The concentration gradient ($C_1 - C_2 = \Delta C$) is the driving force. The unidirectional solute diffusive flux (J_d ; mass removal rate normalized to membrane surface area) through a semipermeable membrane follows Fick's law of diffusion, being directly proportional to the diffusion coefficient

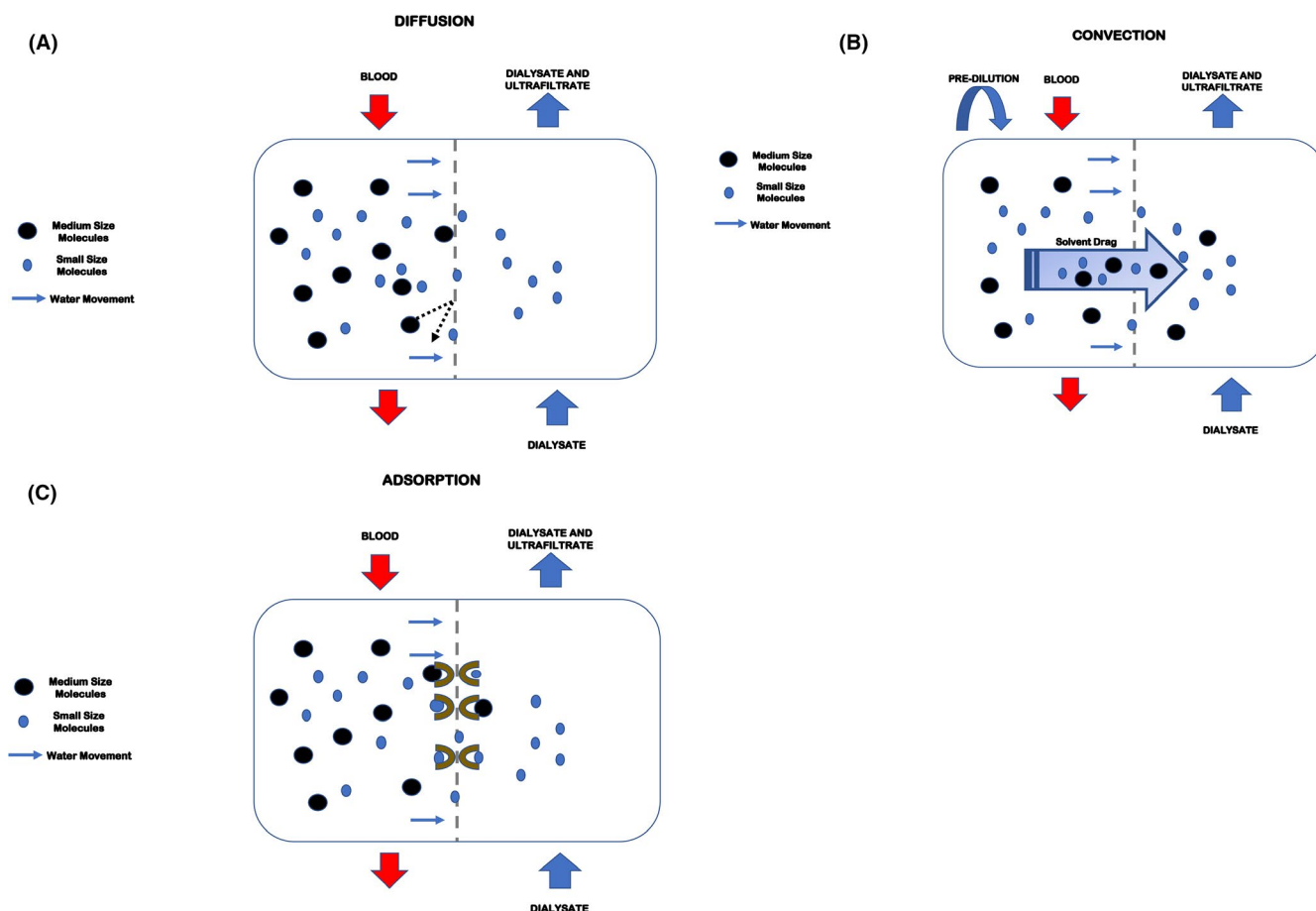


FIGURE 1 Mechanisms of solute removal: Diffusion, convection, and adsorption. (A) Diffusion is the process of transport in which molecules that are present in a solvent and can pass freely across a semipermeable membrane tend to move from the region of higher concentration into the region of lower concentration. (B) Convection is the mass transfer mechanism in which solutes pass across the semipermeable membrane in association with the plasma water ("solvent drag"). (C) Solute in plasma or blood binds to membranes or other extracorporeal removal devices (e.g., resins)

(D) of the solute and inversely proportional to the relevant distance for the components of the system (dx).²

$$J_d = -D \left(\frac{dc}{dx} \right), \quad (2)$$

For CRRT modalities that incorporate diffusion (i.e., CVVHD and continuous veno-venous hemodiafiltration [CVVHDF]), the relevant diffusive components are the blood and dialysate/diafiltrate compartments of the filter along with the membrane. When these three components are assessed in the context of a typical conventional hemodialysis treatment, the blood compartment is rate-limiting for diffusive mass transfer of small solutes while the membrane is limiting for larger solutes. The rationale for using high blood flow rates in this therapy is to minimize “boundary layer” effects at the blood/membrane interface, which account for most of the blood’s resistance to diffusive mass transfer.

However, the flow conditions of CVVHD and CVVHDF create a much different mass transfer environment. For these therapies, as long as a minimal blood flow rate is achieved (typically 100–150 ml/min), dialysate-side mass transfer is actually the rate-limiting component for diffusive removal of small solutes. This is caused by the substantial boundary effects due to the low dialysate flow rates used. In fact, under most CVVHD and CVVHDF operating conditions, dialysate-side mass transfer is so slow that “saturation” of the effluent dialysis (i.e., equivalent concentrations in the plasma water and effluent) is observed for small solutes. For solutes of increasing molecular weight, membrane limitations become more important (due to both pore size restrictions and slow diffusivity in the membrane itself).

Diffusion is an efficient transport mechanism for the removal of relatively small solutes, but as solute molecular weight increases, diffusion becomes limited and the relative importance of convection increases.

3.1.2 | Convection

Convection is the process whereby solutes pass through membrane pores, dragged by fluid movement (ultrafiltration) occurring as a result of a hydrostatic and/or osmotic TMP gradient (Figure 1).³

The sieving coefficient (SC) is the most commonly measured membrane parameter characterizing a filter’s convective solute removal capability. The SC is the ratio of a specific solute concentration in the filtrate (generated under “pure” ultrafiltration operating conditions), divided by the mean plasma concentration in the filter⁴:

$$SC = \frac{C_{UF}}{(C_{pi} + C_{po})/2}, \quad (3)$$

where C_{UF} is the solute concentration in the ultrafiltrate, and C_{pi} and C_{po} the plasma solute concentrations at the inlet and outlet of the

filter, respectively. A true calculation would require measurement of the solute concentration in plasma water rather than plasma to avoid interference of proteins. Nevertheless, for practical purposes, plasma concentration is normally accepted. Because there is little difference between the arterial and venous solute concentrations under CRRT operating conditions, the formula is commonly simplified to the ratio between the concentration in the ultrafiltrate and the concentration in pre-filter plasma:

$$SC = \frac{\text{Ultrafiltrate concentration}}{\text{Blood concentration}}, \quad (4)$$

Sieving coefficient is theoretically measurable only in the absence of a gradient for diffusion (i.e., no concentration gradient through the membrane). However, molecules whose passage through a membrane is restricted to some degree accumulate at the membrane surface as part of a phenomenon called concentration polarization—this process is particularly pronounced for high molecular weight solutes. The accumulation produces a relatively high blood concentration of such solutes in a region just adjacent to the membrane surface (“sub membranous” layer) and results in a concentration gradient across the membrane. Thus, in this case, diffusion supplements convection at least to some extent, even under conditions of “pure” ultrafiltration. For small solutes (e.g., urea and creatinine) that essentially have free passage (i.e., $SC = 1.0$), concentration polarization is not relevant and the assumption of no transmembrane concentration is valid. It should be emphasized that the phenomenon of concentration polarization (specific to a partially rejected solute) is distinct from fouling (non-specific adsorption of many proteins) and the two processes occur simultaneously for membrane-based procedures.

Sieving coefficient is specific for each solute-membrane combination. Regardless of membrane type, all filters at the initial state have small solute SC values of 1, and these values are typically not reported by dialyzer manufacturers. Sieving coefficient values for larger solutes are more applicable and manufacturers frequently provide data for one or more middle molecule surrogates, such as vitamin B12, inulin, and myoglobin. As is the case of solute clearance, the relation between SC and solute molecular weight is highly dependent on membrane pore size.

The convective flux (Jc) of a solute depends on ultrafiltration rate (Q_{UF}), membrane surface area (A), solute concentration in plasma (C_{pi}) and solute SC:

$$Jc = \frac{Q_{UF}}{A \times C_{pi} \times SC}, \quad (5)$$

Compared to diffusive transport, convective transport permits increased removal rates for higher molecular weight solutes.

3.1.3 | Adsorption

As mentioned previously, fouling involves the non-specific adsorption of plasma proteins to a membrane, resulting in decreased effective

permeability. In general, the proteins most importantly contributing to fouling are those found at the highest concentration in the plasma (i.e., albumin, immunoglobulins, fibrinogen, etc)—the size of these proteins precludes pore entry to any significant degree. Moreover such proteins adsorb in relatively small quantities relative to their mass found in the circulating plasma pool—as such, plasma concentrations are not significantly affected. Adsorption of low-molecular proteins (which are normally eliminated by glomerular filtration) may be the dominant or sole mechanism by which some such compounds are removed,⁵ as opposed to a transmembrane mechanism. In turn, the adsorptive process may result in a clinically significant decrease in plasma concentration for certain solute/membrane combinations.

The adsorptive surface area of a membrane resides primarily in the pore structure rather than in the nominal surface area (Figure 1). As such, the adsorption of a low-molecular-weight protein is highly dependent on access of the protein to a membrane's internal pore structure. Consequently, adsorption of peptides and low-molecular-weight proteins (e.g., β_2 -microglobulin) to low-flux membranes is not expected to be clinically significant, at least in comparison with that which occurs to high-flux membranes. The adsorptive affinity of certain high-flux synthetic membranes for proteins and peptides is particularly high, generally attributable to the relative hydrophobicity of these membranes.

3.2 | Fluid removal

3.2.1 | Ultrafiltration

Ultrafiltration describes the transport of plasma water (solvent) through a semipermeable membrane driven by a pressure gradient between the blood and dialysate/ultrafiltrate compartments. As suggested previously, Q_{UF} is a function of TMP and the relationship between these two parameters is established by the effective permeability of the filter membrane at a given time. In turn, this effective permeability is a dynamic parameter which is heavily influenced by the extent of membrane fouling. A decrease in effective permeability requires a concomitant increase in TMP to achieve a specific target Q_{UF} .

It is important to differentiate between Q_{UF} and Q_{NET} . Q_{UF} is the total ultrafiltration rate—this volume can be completely replaced, partially replaced, or not be replaced at all, depending on the patient needs. The net ultrafiltration rate (Q_{NET}) is the difference between Q_{UF} and the rate at which volume is replaced in the circuit, corresponding to the net fluid removal requirements for the patient.

In light of the above, the extent to which ultrafiltration is utilized in CRRT is modality-specific. When only fluid removal is required to control volume overload, slow continuous ultrafiltration (SCUF) can be employed—in this case Q_{UF} and Q_{NET} are equivalent. Convective blood cleansing modalities (continuous veno-venous hemofiltration [CVVH] and CVVHDF) utilize higher Q_{UF} values in conjunction with replacement fluid to both achieve net fluid removal and reductions in blood solute concentrations. Finally, although CVVHD does not involve the use of replacement fluid and thus does not have a

TABLE 1 Different mechanism of solute clearance

Technique	Convection	Difussion
SCUF	+	—
CVVH	++++	—
CVVHD	+	++++
CVVHDF	+++	+++
CVVHFD	++	++++

Abbreviations: —, none; +, negligible; ++, some; +++, marked; +++, major control system; CVVH, continuous veno-venous hemofiltration; CVVHD, continuous veno-venous hemodiafiltration; CVVHFD, continuous veno-venous high-flux dialysis; SCUF, slow continuous ultrafiltration.

convective component, ultrafiltration is still used to achieve net fluid removal. In an analogous manner, the components of the effluent differ according to the modality used. The effluent consists of dialysate and net ultrafiltrate in CVVHD, replacement fluid and net ultrafiltrate in CVVH, and dialysate, replacement fluid, and net ultrafiltrate in CVVHDF.⁶

3.3 | Continuous renal replacement therapy techniques

Several techniques are today available in the spectrum of CRRT. Techniques may differ in terms of vascular access and extracorporeal circuit design, frequency, and intensity of treatment, predominant mechanism of transport used, and type of membrane (Table 1).

3.3.1 | Slow continuous ultrafiltration

Slow continuous ultrafiltration typically is employed for 24 h a day but also may be applied during some portion of the day. The objective is to achieve volume control in patients with severe, diuretic-resistant volume overload. Extracorporeal ultrafiltration is being used increasingly as an adjunctive therapy for patients with refractory heart failure.^{7,8} Ultrafiltration rates typically employed in SCUF (<10 ml/min) are significantly less than those in hemofiltration, which may be 40 ml/min or higher. Therefore, although the minimum blood flow rate may be 200 ml/min or higher in the setting of post-dilution hemofiltration, a blood flow rate of 50 ml/min in SCUF may be adequate to maintain the filtration fraction less than 25% (see below). Although very effective for volume reduction, the low filtration rates and lack of replacement fluids render this therapy ineffective as a blood purification modality.

3.3.2 | Continuous veno-venous hemofiltration

As discussed previously, hemofiltration is based on the mechanism of convection, which involves the mass transfer of solutes in

association with ultrafiltered plasma water ("solvent drag"). Due to the limited ability of diffusion to eliminate compounds beyond the small solute range, the primary rationale for the use of CVVH is to achieve an extended molecular weight spectrum of solute removal.

Replacement fluids play an integral part in the delivery of CVVH (and CVVHDF). As indicated above, a Q_{UF} value greater than that required simply for net fluid removal is used to increase convective solute clearance. As solutes are eliminated in the effluent, the concentrations of solutes remaining in the blood are reduced by the simultaneous administration of replacement fluid, which effectively dilutes the remaining solute. This exchange process achieves a reduction in blood concentrations of solutes that are substantially eliminated in the effluent (i.e., those having relatively high SCs). Due to the large difference between Q_{UF} and Q_{NET} , severe volume depletion would occur rapidly without the use of replacement fluid. Thus, the other major rationale for the use of replacement fluid is to maintain desired fluid balance.

Replacement fluids can be delivered to the extracorporeal circuit either before the filter ("pre-dilution") or after the filter ("post-dilution"). There are advantages and disadvantages of each approach. Based on the relationship between solute clearance and required replacement fluid rate, the post-dilution mode theoretically provides the most efficient way to achieve solute removal. In this mode, the concentrations of solute in the blood delivered to the filter are the same as plasma concentrations. However, post-dilution hemofiltration is limited inherently by the attainable blood flow rate and the associated filtration fraction maximum, which is generally considered to be 25%. (The classical definition of filtration fraction is the ratio of ultrafiltration rate to plasma flow rate delivered to the filter.) Operating a filter at or near this filtration fraction limit increases the chance not only of filter clotting but also pore blockage due to the high concentrations of proteins comprising the secondary membrane at the membrane surface ("clogging"). The blood flow rate limitations imposed by the use of temporary catheters accentuates the filtration fraction-related constraints on maximally attainable ultrafiltration rate in the post-dilution mode.

From a mass transfer perspective, the use of pre-dilution has several potential advantages over post-dilution.⁹ Hematocrit and blood total protein concentration are reduced significantly before the entry of blood into the hemofilter. This effective reduction in the red cell and protein content of the blood reduces the risk of fouling, concentration polarization and clotting, resulting in improved mass transfer. Pre-dilution also favorably affects mass transfer because of augmented flow in the blood compartment, as pre-filter mixing of blood and replacement fluid occurs. This achieves a relatively high membrane shear rate, which also reduces solute-membrane interactions. Finally, pre-dilution also may enhance mass transfer for some compounds by creating concentration gradients that induce solute movement out of red blood cells. However, because the clearance of solutes is dependent on their concentration in the filter, the major drawback of pre-dilution hemofiltration is its relatively low efficiency related to dilution. This results in relatively high replacement

fluid requirements in the pre-dilution mode to achieve a given solute clearance.¹⁰

3.3.3 | Continuous veno-venous hemodialysis

Continuous hemodialysis first was proposed by Scribner in the early 1960s with an eye toward renal replacement for end stage kidney disease (ESKD).^{11,12} When continuous hemodialysis was initially used for AKI in the mid-1980s, the arteriovenous mode, in which blood flow through the circuit is driven by the patient's mean arterial pressure rather than a pump, was utilized. (The arteriovenous mode was also used in the earliest application of hemofiltration for AKI). The arteriovenous techniques have been replaced by veno-venous systems, in which a blood pump provides more reliable operation and solute clearance.^{13,14} CVVHD is characterized by slow countercurrent dialysate flow into the ultrafiltrate/dialysate compartment of the dialyzer. The prevalent mechanism of solute transport in this technique is diffusion, with the prescribed ultrafiltration rate targeted to achieve the patient's desired fluid balance (i.e., without requirement of replacement fluids).

Over the past two decades, both the surface area of filters used for CVVHD (along with CVVH) and dialysate flow rates have progressively increased in parallel with higher CRRT dose targets. Although solute removal in CVVHD is limited by slow dialysate-side mass transfer (as mentioned previously), membrane surface area is also important, especially as dialysate flow rate increases. Clinical studies indicate filter surface areas of 1.0 m² or higher are generally required to preserve dialysate saturation with respect to small solutes.

Comparisons of CVVHD and CVVH using similar membranes have shown little difference in solute clearance for small and middle-molecular weight solutes. CVVHD is slightly more efficient than pre-dilution CVVH at eliminating small-molecular weight solutes but similar in efficiency to post-dilution CVVH.¹⁵ However, β_2 -microglobulin is cleared more effectively with CVVH due to convection's greater clearance of higher-molecular-weight solutes.^{16,17}

A variant of CVVHD involves the use of a large-pore filter (rather than a standard high-flux filter) to extend the solute removal spectrum of CVVHD to approximate (or even extend beyond) what can be achieved with CVVH. The average pore size of this type of "super high-flux" membrane allows passage of appreciable amounts of albumin—as such, its use cannot be recommended in the CVVH mode. These filters can achieve substantial removal of myoglobin and cytokines, although the clinical implications of extracorporeal mediator removal remain to be determined.

3.3.4 | Continuous veno-venous hemodiafiltration

Continuous veno-venous hemodiafiltration operates combining the principles of hemodialysis and hemofiltration. Dialysate is circulated countercurrent to blood and, at the same time, ultrafiltration is

BOX 1 Solute clearance in continuous renal replacement therapies**CVVHD / CVVHDF**

$$K = E \times Q_D$$

(E = Concentration in effluent dialysate or diafiltrate / Concentration in blood)

Post-dilution CVVH

$$K = SC \times Q_{UF}$$

(SC = Concentration in filtrate / Concentration in blood)

Predilution CVVH

$$K = SC \times Q_{UF} \times (Q_P / Q_P + Q_R)$$

obtained in excess of the desired net fluid loss from the patient. The ultrafiltrate is replaced to some extent with reinfusion fluid, either in pre-dilution or post-dilution mode. Later generation CRRT machines allow a combination of predilution and post-dilution with the aim of combining the advantages of both reinfusion techniques. For a specific solute, the balance between diffusive and convective solute removal is dictated by the specific set of CVVHDF operating conditions, namely blood flow rate, dialysate flow rate, ultrafiltration rate, and filter type. Some investigators suggest CVVHDF is the optimal modality to achieve the broadest spectrum of solute molecular-weight range in the most efficient way.

The specific manner in which diffusion and convection interact in CVVHDF differs significantly from the situation when this treatment is applied in the setting of intermittent hemodialysis for end-stage kidney disease. In the latter situation, diffusion and convection interact in such a manner that total solute removal is significantly less than what is expected if the individual components are simply added together.¹⁸ Diffusive solute removal results in a decrease in solute concentration in the blood compartment of the filter along the axial length (i.e., from blood inlet to blood outlet) of the filter. Because convective solute removal is directly proportional to the blood compartment concentration, convective solute removal decreases as a function of this axial concentration gradient. At the same time, hemoconcentration resulting from ultrafiltration of plasma water causes a progressive increase in plasma protein concentration and hematocrit along the axial length of the filter. This hemoconcentration and resultant hyperviscosity causes an increase in diffusive mass transfer resistance and a decrease in solute transport by this mechanism.¹⁹

Because of the markedly lower flow rates used and clearances obtained in CVVHDF, the effect of simultaneous diffusion and convection on overall solute removal is different. Therefore, the small-solute concentration gradient along the axial length of the filter (i.e., extraction) is minimal compared with that which is seen in chronic hemodialysis or hemodiafiltration, in which extraction ratios of 50% or more are the norm. Thus, the minimal diffusion related change in small-solute concentrations along the filter length allows

any additional clearance related to convection to be simply additive to the diffusive component. This has been demonstrated clearly in CVVHD and CVVHDF.^{20,21}

3.4 | Quantification of solute removal in CRRT**3.4.1 | Clearance**

Understanding quantification of solute removal in CRRT is complicated by the confusion relating to the relationship between clearance and mass removal for different renal replacement therapies. By definition, solute clearance (K) is the ratio of mass removal rate (N) to blood solute concentration (C_B):

$$K = \frac{N}{C_B}, \quad (6)$$

In CRRT, the mass removal rate is typically estimated by measuring the actual mass of solute appearing in the effluent. The mass removal rate is the product of the effluent flow rate (Q_{Eff}) and the effluent concentration of the solute (C_{Eff}).

In CVVHD and CVVHDF, the effluent is dialysate and diafiltrate, respectively (plus net ultrafiltrate related to fluid removal requirements). For these therapies, the extent of solute extraction from the blood is estimated by the equilibration ratio (E), also known as the degree of effluent saturation. The benchmark for efficiency in these therapies is the volume of fluid (dialysate [Q_D] and/or replacement fluid [Q_R]) required to achieve a certain solute clearance target.

Clearance in post dilution continuous veno-venous hemofiltration (CVVH) is the product of the SC and the ultrafiltration rate (Q_{UF}). For small solutes like urea and creatinine, the SC is essentially 1 (assuming a properly functioning filter). Therefore, small solute clearance in post-dilution CVVH essentially is equal to the Q_{UF} . Estimation of clearance in the pre-dilution CVVH has to account for the fact that the blood solute concentrations are reduced by dilution of the blood before it enters the filter. Therefore, the clearance has a

"dilution factor" which essentially is the ratio of the blood water flow rate (Q_p) to the sum of Q_p and the replacement fluid rate (Q_R). The dilution factor can be viewed as a measure of the extent to which predilution differs from post dilution for a specific combination of Q_B and Q_{UF} . (Box 1).

3.4.2 | Filtration fraction

As mentioned previously, filtration fraction is defined as the ratio between the ultrafiltration flow rate (Q_{UF}) and the plasma flow rate (Q_p), and it is used to quantify the phenomenon of hemoconcentration.

$$\text{Filtration fraction} = \frac{Q_{UF}}{Q_p}. \quad (7)$$

In general, a maximal filtration fraction of 25% usually guides prescription in post-dilution CVVH, which is the relevant benchmark in this instance. At filtration fractions beyond this value, hemoconcentration is associated with an environment that promotes interactions between formed elements and proteins in the blood and the filter membrane, leading to a high risk of compromised filter performance resulting from widespread pore occlusion.

Two caveats must be discussed. The plasma flow rate component of the filtration fraction calculation is often based on the hematocrit value at the start of therapy. However, as ultrafiltration proceeds and net volume removal occurs, hematocrit increases. Therefore, for a given volume of blood flowing through the filter, an increasing percentage of that volume is composed of red blood cells and a decreasing percentage is composed of plasma during ongoing ultrafiltration-based therapy. At a fixed blood flow rate and ultrafiltration rate, this implies an increasing filtration fraction, because plasma flow rate is the denominator in the filtration fraction equation. Thus, from the relatively narrow perspective of filtration fraction, a seemingly adequate blood flow rate at the onset of ultrafiltration may be inadequate after long duration of therapy.

A second important consideration related to blood flow rate involves its effect on blood rheology at the membrane surface. The shear rate that blood achieves while passing through an individual hollow fiber membrane is directly proportional to its blood flow rate.²² As indicated previously, higher shear rates have a mitigating effect on fouling, thus potentially preserving membrane permeability to some extent.

3.5 | Dose definition in CRRT

It is generally accepted by clinicians that CRRT affords greater solute removal capabilities relative to intermittent therapies are generally recognized by clinicians. However, approaches based on quantifying

clearance of a specific solute are not used for CRRT dose estimation. This is in contrast to chronic hemodialysis, in which urea kinetic modeling is widely employed. Instead, the landmark trial performed by Ronco et al. has established effluent-based dosing as the standard of care for CRRT.²³

The foundation for this approach is the assumed correlation (not equivalence) between normalized effluent rate (mL/kg/hr) and small solute clearance. However, this correlation differs across CRRT modalities and the specific operating conditions used for a specific modality. Nevertheless, the 17th Acute Disease Quality Initiative consensus meeting has validated effluent flow rate as an acceptable surrogate for CRRT dose.²⁴

The major factors causing divergence between the effluent-based dose and actual small solute clearance are impaired filter performance, treatment downtime during the course of a particular day, and pre-dilution.¹⁹ Filter performance may degrade over time with fouling and filter clotting as the most important causative mechanisms.²⁵ While the effect of fouling is most pronounced on larger solutes under convective conditions, it can influence the clearance of a solutes, including urea and creatinine. As stated previously, the use of higher blood flow rates may mitigate fouling to some extent. Clotting leads to loss of effective membrane surface area as a certain percentage of fibers become inaccessible to blood flow. While adequate anticoagulation of the extracorporeal circuit can reduce the possibility of filter clotting, it has no impact on fouling. The influence of pre-dilution on solute clearance in convective modalities is determined by the relationship between blood flow rate and replacement fluid rate. When relatively low blood flow rates (150 mL/min or less) are prescribed for therapies aiming to achieve delivered doses of 25 mL/kg/h or more, the pre-dilution "penalty" typically falls in the 15%–20% range for CVVHDF and 30% or more for CVVH.²⁶

Continuous renal replacement therapy dose prescription, delivery and solute control are a dynamic process. Prescribed dose can be modified according to patient demand and in response to iterative evaluation of quality measures like effective treatment time, clearance, and the ratio of delivered/prescribed dose. Prescribed dose should be evaluated at least once every 24 h and more often according to patient needs.²⁴

Finally, clearance of urea and middle molecular weight molecules may not be the major determinants of short-term outcomes in patients with AKI.²⁷ As suggested by Davenport and Farrington, the adequate removal of what they call "very small waste products" (potassium, sodium, and hydrogen ions) and correction of fluid overload are the main determinants of RRT adequacy in patients with AKI since the consequences of their accumulation could be lethal in hours to days.²⁷ Therefore, fluid management is an integral component for managing critically ill patients with AKI to correct abnormalities in plasma composition and maintain fluid balance. The composition and site of administration of CRRT fluids are varied to achieve specific parameters of solute, electrolyte and acid-base balance required for organ support. The volume of fluid removed and infused can be manipulated to provide precise control for fluid balance and solute control.²⁸

4 | CONCLUSION

Rational prescription of CRRT to critically ill patients with AKI needs to be based on an understanding of the basic principles of solute and water removal. CRRT employs a variety of mechanisms, including ultrafiltration, diffusion, convection, and adsorption to achieve solute and water elimination. These principles of solute and water removal are applied in a therapeutic context of the various CRRT modalities used for treating patients with AKI.

By removing solutes through convection, diffusion, and adsorption and replenishing depleted solutes selectively by varying the composition of replacement solution and/or dialysate, CRRT can be used to achieve solute balance to a desired level. Understanding the basic mechanisms underlying the process of CRRT is essential to be able to make appropriate treatment choices for individual patients.

CONFLICT OF INTEREST

The authors declare no potential conflicts of interest in relation with the content of this manuscript.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

ORCID

Rolando Claure-Del Granado  <https://orcid.org/0000-0002-8163-3779>

REFERENCES

1. Uchino S, Kellum JA, Bellomo R, et al. Acute renal failure in critically ill patients: a multinational, multicenter study. *JAMA*. 2005;294(7):813-818.
2. Clark WR, Hamburger RJ, Lysaght MJ. Effect of membrane composition and structure on solute removal and biocompatibility in hemodialysis. *Kidney Int*. 1999;56(6):2005-2015.
3. Lee K, Jeong JH, Mun CH, et al. Convection-enhanced high-flux hemodialysis. *Artif Organs*. 2007;31(8):653-658.
4. Neri M, Villa G, Garzotto F, et al. Nomenclature for renal replacement therapy in acute kidney injury: basic principles. *Crit Care*. 2016;20(1):318.
5. Clark WR, Macias WL, Molitoris BA, Wang NH. Membrane adsorption of beta 2-microglobulin: equilibrium and kinetic characterization. *Kidney Int*. 1994;46(4):1140-1146.
6. Ficheux A, Ronco C, Brunet P, Argiles A. The ultrafiltration coefficient: this old 'grand inconnu' in dialysis. *Nephrol Dial Transplant*. 2015;30(2):204-208.
7. Costanzo MR, Saltzberg M, O'Sullivan J, Sobotka P. Early ultrafiltration in patients with decompensated heart failure and diuretic resistance. *J Am Coll Cardiol*. 2005;46(11):2047-2051.
8. Clark WR, Paganini E, Weinstein D, Bartlett R, Sheinfeld G, Ronco C. Extracorporeal ultrafiltration for acute exacerbations of chronic heart failure: report from the Acute Dialysis Quality Initiative. *Int J Artif Organs*. 2005;28(5):466-476.
9. Huang Z, Li B, Zhang W, Gao D, Kraus MA, Clark WR. Convective renal replacement therapies for acute renal failure and end-stage renal disease. *Hemodial Int*. 2004;8(4):386-393.
10. Clark WR, Turk JE, Kraus MA, Gao D. Dose determinants in continuous renal replacement therapy. *Artif Organs*. 2003;27(9):815-820.
11. Scribner BH, Caner JE, Buri R, Quinton W. The technique of continuous hemodialysis. *Trans Am Soc Artif Intern Organs*. 1960;6:88-103.
12. Geronemus R, Schneider N. Continuous arteriovenous hemodialysis: a new modality for treatment of acute renal failure. *Trans Am Soc Artif Intern Organs*. 1984;30:610-613.
13. Ronco C, Bellomo R. Basic mechanisms and definitions for continuous renal replacement therapies. *Int J Artif Organs*. 1996;19(2):95-99.
14. Ronco C, Bellomo R. Continuous renal replacement therapy: evolution in technology and current nomenclature. *Kidney Int Suppl*. 1998;66:S160-S164.
15. Parakinkas D, Greenbaum LA. Comparison of solute clearance in three modes of continuous renal replacement therapy. *Pediatr Crit Care Med*. 2004;5(3):269-274.
16. Wynckel A, Cornillet J, Béné B, et al. Improved removal of small proteins using continuous venovenous hemofiltration to treat acute renal failure. *ASAIO J*. 2004;50(1):81-84.
17. Ricci Z, Ronco C, Bachetoni A, et al. Solute removal during continuous renal replacement therapy in critically ill patients: convection versus diffusion. *Crit Care*. 2006;10(2):R67.
18. Jaffrin MY. Convective mass transfer in hemodialysis. *Artif Organs*. 1995;19(11):1162-1171.
19. Claure-Del Granado R, Mehta RL. Assessing and delivering dialysis dose in acute kidney injury. *Semin Dial*. 2011;24(2):157-163.
20. Sigler MH, Teehan BP. Solute transport in continuous hemodialysis: a new treatment for acute renal failure. *Kidney Int*. 1987;32(4):562-571.
21. Brunet S, Leblanc M, Geadah D, Parent D, Courteau S, Cardinal J. Diffusive and convective solute clearances during continuous renal replacement therapy at various dialysate and ultrafiltration flow rates. *Am J Kidney Dis*. 1999;34(3):486-492.
22. Clark WR. Quantitative characterization of hemodialyzer solute and water transport. *Semin Dial*. 2001;14(1):32-36.
23. Ronco C, Bellomo R, Homel P, et al. Effects of different doses in continuous veno-venous haemofiltration on outcomes of acute renal failure: a prospective randomised trial. *Lancet*. 2000;356(9223):26-30.
24. Bagshaw SM, Chakravarthi MR, Ricci Z, et al. Precision continuous renal replacement therapy and solute control. *Blood Purif*. 2016;42(3):238-247.
25. Macedo E, Claure-Del Granado R, Mehta RL. Effluent volume and dialysis dose in CRRT: time for reappraisal. *Nat Rev Nephrol*. 2011;8(1):57-60.
26. Claure-Del Granado R, Macedo E, Chertow GM, et al. Effluent volume in continuous renal replacement therapy overestimates the delivered dose of dialysis. *Clin J Am Soc Nephrol*. 2011;6(3):467-475.
27. Davenport A, Farrington K. Dialysis dose in acute kidney injury and chronic dialysis. *Lancet*. 2010;375(9716):705-706.
28. Murugan R, Hoste E, Mehta RL, et al. Precision fluid management in continuous renal replacement therapy. *Blood Purif*. 2016;42(3):266-278.

How to cite this article: Claure-Del Granado R, Clark WR. Continuous renal replacement therapy principles. *Semin Dial*. 2021;34(6):398-405. <https://doi.org/10.1111/sdi.12967>